HoNE: Hold-out and Natural-Novelty Evaluation of Learned NIDS for Zero-Day Detection

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***Abstract*—Operators deploy learned network intrusion detection systems (NIDS) to catch attacks that did not exist at training time, yet zero-day claims usually rest on leave-one-attack-class-out (LOACO) evaluation on a random split, scored once and summarised by AUC. We introduce HoNE (Hold-out and Natural-novelty Evaluation), a pre-registered, leakage-controlled benchmark that scores the same detectors, data and seeds under LOACO and under natural novelty, where attack classes first appear later in time, including four rolling cut-points (45 class–cut units, 19 attack classes). We evaluate four NetFlow-v3 datasets and Lycos2017 with 12 detectors on a recurrent backbone, replicated on a GRU backbone, over five seeds. Five findings emerge. Temporal LOACO is infeasible on all three day-scheduled datasets. The winning detector family changes with protocol, dataset and cut-point: XGBoost matches the best novelty family in four of seven fixed evaluations but trails benign-only detectors over the rolling cuts. Natural novelty raises the seed variability of the first backbone's scores by 1.7–17×, and of the GRU's representation scores by 2.6×. Max-softmax and energy fall below chance on both backbones. AUCs of 0.53–0.75 mean at most 18% of novel attacks caught at 1% false-positive rate. HoNE turns these findings into eight pre-deployment checks.**

***Index Terms*—network intrusion detection, network security management, zero-day attacks, evaluation methodology, NetFlow**

I. Introduction

Network operators increasingly rely on machine-learning network intrusion detection systems (NIDS) because signature rules cannot cover attacks that did not exist when the rules were written. The value of such a detector lies almost entirely in its *zero-day* behaviour: an attack it has already seen could equally be caught by a signature. Before a learned NIDS is trusted in production, therefore, someone must decide whether its zero-day claim is credible, and that decision usually rests on evidence supplied by the detector's developers.

The stakes of that decision are operational. A detector whose zero-day performance has been overstated misses the attacks it was bought to catch, while one whose performance varies from training run to training run gives the security team no stable basis for setting alert thresholds or staffing triage. Both failures surface only after deployment, when the cost of discovering them is highest. The quality of pre-deployment evidence is therefore a network-management question, not only a machine-learning one.

The evidence in current use has a remarkably uniform shape, and each of its parts is fragile. Zero-day ability is measured by *leave-one-attack-class-out* (LOACO): one attack class is removed from training and the detector must flag it at test time, on a benchmark split at random [1], [2]. Random splits let near-identical flows and whole attack episodes appear in both training and test sets [3], [4], and LOACO keeps the benign traffic of the test period in training, so the held-out class is the only thing that changes. Results usually come from one training run, and novelty is often scored with confidence measures such as maximum softmax probability [5] or energy [6], which were designed for multi-class image recognition. The security-ML literature has long warned that such benchmark scores overstate field performance [7], [8].

What has not been tested is whether LOACO evidence predicts the situation an operator actually faces: a new attack class that arrives *later in time*, alongside drifting benign traffic, which we call *natural temporal novelty*. Answering this requires scoring the same detectors, on the same data and seeds, under both protocols, with leakage controlled and run-to-run variability reported. No existing study does all of this together, so it remains unknown whether a detector that wins under LOACO would still win, or even remain stable, under natural novelty.

To close this gap we introduce **HoNE** (**Ho**ld-out and **N**atural-novelty **E**valuation), an evaluation benchmark rather than a new detector. HoNE has four components (Fig. 1): leakage-controlled splits built from flow timestamps; paired zero-day protocols that evaluate every detector under both LOACO and natural novelty; a panel of 12 detectors in five families, replicated on a second backbone and each reported with its seed variability and its detection rate at an operational false-positive rate; and an acceptance protocol that turns the benchmark's findings into checks an operator can require. The benchmark was pre-registered: hypotheses, metrics and gates were hashed before every experiment, and every amendment is published.

Applying HoNE to five public datasets yields five findings, each stated as an operator concern:

- **The standard test is often impossible on realistic captures.** Temporal LOACO yields zero feasible folds on CIC-IDS2018, ToN-IoT and Lycos2017, while natural novelty offers 2–5 evaluable classes per split and 45 class–cut units over four rolling cut-points (Sec. IV-B).
- **The protocol picks the winner.** The leading detector family changes with dataset, protocol and cut-point, identically on both backbones. A gradient-boosted tree matches or beats the best novelty family in four of seven fixed evaluations, but trails benign-only detectors over the rolling cuts (Sec. IV-C).

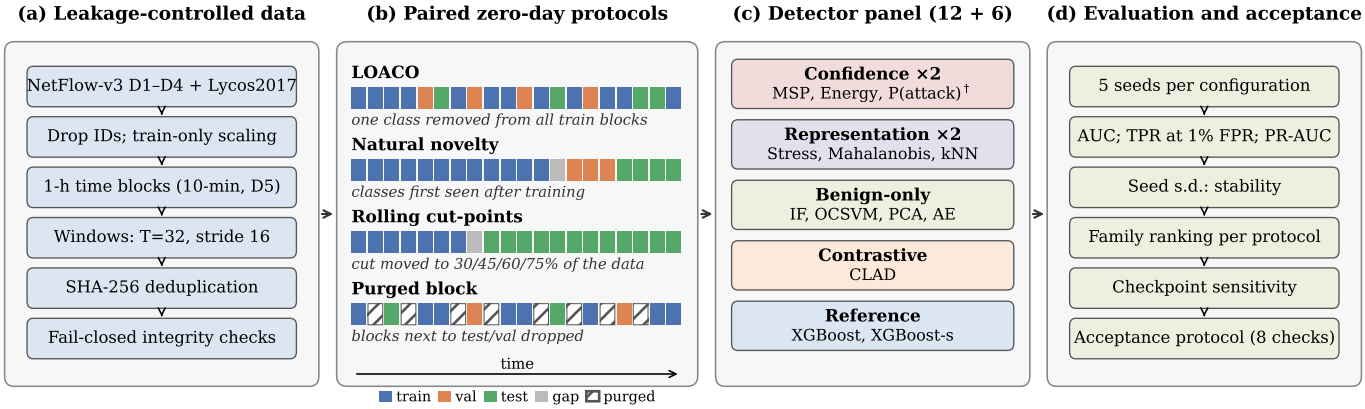


Fig. 1. HoNE overview. (a) Flows are grouped into time blocks before windowing and deduplicated, and every build runs fail-closed integrity checks. (b) Each detector is evaluated under class hold-out (LOACO) and natural temporal novelty, plus a purged-block split for short captures. (c) Twelve detectors in five families are scored on the same windows (†: added after the final gate). (d) Results are reported per class over five seeds, with seed variability and family rankings, and condensed into the HoNE acceptance protocol.

- **One training run is not evidence.** Natural novelty raises the seed variability of the first backbone's scores by 1.7–17× relative to LOACO and of the GRU's representation scores by 2.6× (Sec. IV-D).
- **Some scores are invalid by construction.** Max-softmax and energy fall below chance in binary NIDS on both backbones, while the attack probability does not (Sec. IV-E).
- **AUC overstates zero-day coverage.** Over the rolling cuts no family catches more than 18% of novel attack windows at 1% false-positive rate, although family AUCs reach 0.75 (Sec. IV-G).
- **Shift effects are dataset-specific.** Temporal splits degrade one dataset severely and leave others intact, and zero-shot transfer is near chance (Sec. IV-F).

Sec. II reviews existing approaches and the gap they leave. Sec. III derives HoNE's requirements and design, Sec. IV reports the evaluation, Sec. V interprets the findings and presents the acceptance protocol, and Sec. VI concludes.

II. Related Work

Prior work on learned NIDS evaluation falls into four lines: studies of evaluation pitfalls, zero-day and open-set detectors, cross-domain studies, and sequence-based detectors. Each contributes a piece of what an operator needs, and each leaves a different piece missing.

Evaluation pitfalls and datasets. Arp et al. catalogued ten pitfalls in security ML, including data snooping and inappropriate baselines [3]; Apruzzese et al. showed that ML-NIDS evaluations rarely reflect operational assumptions [8]; and dataset audits found flow-construction and labelling errors in CIC-IDS2017 [4], [9] and inconsistencies across dataset-generation tools [10]. The NetFlow standardization of Sarhan et al. made datasets comparable [11], [12], and its third version adds per-flow timestamps [13]. These works diagnose *how* evaluations go wrong, but they do not compare zero-day protocols or measure what the pitfalls do to detector rankings.

Zero-day and open-set detection. Zoppi et al. found that unsupervised detectors handle simulated unknown attacks better than supervised ones [14], and Sarhan et al. linked low zero-day detection rates to distribution distance [15]. Recent detectors instead train with known attacks: CLAD learns benign structure with a contrastive loss [1], and Jodayree et al. combine a transformer with energy scoring and extreme-value calibration [2]. These studies measure zero-day ability, but all of them use class hold-out, usually without reporting seed variability, so their conclusions inherit LOACO's assumptions.

Cross-domain generalization. Layeghy and Portmann [16], DI-NIDS [17] and recent work on flow-based IoT detection [18] show that detectors degrade sharply on networks other than the one they were trained on. This line addresses shift *between* networks, whereas an operator's first concern is shift *over time within* the network being protected.

Research gap and positioning. Taken together, these lines leave one question unanswered: does the hold-out evidence used to support zero-day claims agree with how detectors behave under natural temporal novelty, on the same detectors, data and seeds? Answering it requires leakage control, paired protocols, a broad detector panel and variability reporting at once. HoNE provides all four, and it differs from prior benchmarks in treating the evaluation protocol itself as the object of study. Appendix A extends this review to sequence, neuromorphic and drift-aware NIDS and positions HoNE against prior work in Table VIII.

III. Methodology

This section first derives the requirements that a trustworthy zero-day evaluation must meet, then describes how each HoNE component implements one of them.

A. Requirements

The gap in Sec. II translates into four requirements. **R1, leakage control:** near-duplicate windows and attack episodes must not span training and test sets. **R2, protocol pairing:**

TABLE I
DATASETS. WINDOWS ARE COUNTED BEFORE DEDUPLICATION.

ID	Dataset	Flows	Windows	Dupl.	Classes
D1	NF-CIC-IDS2018-v3	20.1 M	1,257,018	1,838	14 (6 fam.)
D2	NF-UNSW-NB15-v3	2.37 M	147,801	0	9
D3	NF-ToN-IoT-v3	27.5 M	1,719,860	166,669	9
D4	NF-BoT-IoT-v3	16.9 M	890,852[a]	–	4
D5	Lycos2017	1.84 M	114,783	0	13

[a]Windows retained; D4 has no split of its own (transfer target only).

TABLE II
DETECTORS (12) IN FIVE FAMILIES. CONFIDENCE AND REPRESENTATION SCORES ARE ALSO COMPUTED ON A GRU BACKBONE; XGBOOST-S IS A SEEDED VARIANT.

Family	Detectors	Fitted on
Confidence	MSP [5], Energy [6], $P(\text{attack})^\dagger$	Backbone output
Representation	Stress, Mahalanobis [25], kNN [26]	Backbone features
Benign-only	IF [27], OCSVM [28], PCA, autoencoder	Benign windows
Contrastive	CLAD [1]	Benign + attacks
Reference	XGBoost [29], XGBoost-s	Labelled windows

every detector must be scored under both LOACO and natural temporal novelty on the same data and seeds, and infeasible protocols must be reported rather than silently skipped. **R3, score breadth and validity:** detectors from different families, including simple baselines, must be compared, and each score must be meaningful for a binary detector. **R4, variability and accountability:** results must be reported over several seeds with per-class variability, and protocol changes must be recorded. HoNE's four components map one-to-one onto these requirements (Fig. 1a–d).

B. Leakage-controlled data (R1)

D1–D4 are the NetFlow-v3 versions of CSE-CIC-IDS2018 [19], UNSW-NB15 [20], ToN-IoT [21] and BoT-IoT [22], which share 53 features [13]; D5 is Lycos2017, a corrected re-extraction of CIC-IDS2017 [9], processed in its own feature space (Table I). D4 contains only 1,315 benign windows and serves only as a transfer target, and D1's 14 file labels are grouped into six attack families so that a held-out class has no near-identical sibling in training.

To meet R1, addresses, ports and DNS query IDs are dropped, flow timestamps are used only for ordering and splitting, and TTL fields are dropped as a known UNSW-NB15 label-leakage artifact. Infinite values are imputed with the training median, heavy-tailed counts are $\log(1+x)$-transformed, and features are standardized with training statistics, leaving 41 common features over D1–D3. Flows are then sorted by start time and grouped into 1-hour blocks (10-minute blocks for D5's purged split) *before* windowing into $T{=}32$ flows with stride 16, so that no window crosses a block. A window is an attack if any flow is, and its class is the majority attack class. Windows are hashed and exact duplicates removed globally, which removed 9.7% of D3's windows.

Because a silent leakage bug would invalidate every later result, each build runs fail-closed integrity checks covering hash- and block-level disjointness of splits, class ratios, the absence of held-out flows from LOACO training sets, and finite values. All passed, including 538 checks on the D1–D4 archives after Amendment 1 and 32 on the D5 purged-block build.

C. Paired zero-day protocols (R2)

On these leakage-controlled windows, HoNE defines two splits and three protocols. *grouped_random* shuffles blocks 60/20/20 into training, validation and test sets, and *temporal_gap* orders them in time, with the first 60% for training, one discarded gap block, then 20% each for validation and test. *LOACO* runs on grouped_random: a class is eligible if its training pool holds at least 200 windows (by majority label), and it is evaluated only if it also occurs in the test split. *Natural novelty* runs on temporal_gap and scores the classes that occur in test but never in training; its validation set is the last 20% of the training period, because otherwise some datasets have attacks in every validation window. For short captures, the *purged-block* split assigns 10-minute blocks 60/20/20 within each capture day and removes blocks adjacent to test or validation blocks, following purged cross-validation [23]; its folds additionally require 50 test windows. *Rolling cut-points* (Amendment 7) re-apply the natural-novelty rule with the training period ending at 30, 45, 60 and 75% of each dataset's windows, one discarded gap block, and testing on everything after; the cleaner is refitted on each cut's training rows, and each (dataset, cut, class) is one unit. Training sets are balanced 1:1 (7,000 windows per class), and validation and test sets use a 5:1 benign-to-attack ratio.

D. Detector panel (R3)

Meeting R3 requires detectors that read different evidence, so HoNE groups 12 scores into five families by what they read from a model [24] (Table II). *Confidence* scores read the classifier output; *representation-distance* scores read hidden features [25], [26]; *benign-only* detectors never see attacks [27], [28]; the *contrastive* detector CLAD [1] is trained on benign and known attacks; and XGBoost [29] on flattened windows is a supervised reference. Higher scores always mean more anomalous, and orientation was fixed before any run.

Confidence and representation scores come from a recurrent backbone (AHSD) built around a fixed benign reference E. Each flow vector is embedded as $z_t = \mathrm{LayerNorm}(Wx_t) \in \mathbb{R}^{128}$, and a state h_t, initialised at E, tracks the input relative to it with per-step, per-channel gates:

$$h_t = h_{t-1} + \alpha_t \odot (z_t - E) - \beta_t \odot (h_{t-1} - E), \qquad (1)$$

with gates $[\alpha_t, \beta_t] = \sigma(W_g[z_t, h_{t-1}])$. E is not learned: it is the mean embedding of benign training windows, tracked by an exponential moving average during training and set exactly afterwards. Stress $= \frac{1}{TD}\sum_{t,d}|h_{t,d} - E_d|$, and a two-layer head (linear–GELU–linear) reads $[h_T,\ \frac{1}{T}\sum_t h_t,\ \text{stress}]$ and outputs $P(\text{attack})$; its hidden layer is the representation used by Mahalanobis and kNN. To test whether findings depend on this backbone, every confidence and representation score

is recomputed on a second backbone, a one-layer GRU (hidden size 128) with the same embedding and head (without stress), trained identically (Amendment 7). XGBoost uses subsample and column sampling of 1 and is therefore deterministic; XGBoost-s uses 0.8 for both with the run's seed, so that the stability comparison is not made against a deterministic baseline.

The validity part of R3 matters most for the confidence family. In a two-class detector with logit margin $\Delta = f_{\text{attack}} - f_{\text{benign}}$, MSP $= \max(P(\text{attack}), 1 - P(\text{attack})) = \sigma(|\Delta|)$ measures how *sure* the model is, not *which* class it chose; temperature calibration rescales Δ and cannot change this ranking. A new attack that the classifier detects confidently has MSP close to 1, so the anomaly score $1 - \text{MSP}$ is close to 0 and the attack reads as in-distribution; energy behaves the same way because it decreases as the largest logit grows. HoNE therefore reports $P(\text{attack})$ alongside MSP and energy; because it was added after the final pre-registered gate (marked †), comparisons involving it are diagnostic rather than confirmatory.

E. Variability and accountability (R4)

Every configuration is run with five seeds, and results are reported per class with their seed standard deviation, so that run-to-run variation is visible rather than averaged away. Besides ROC AUC we report the true-positive rate at 1% false-positive rate (TPR@1%), the quantity a security team experiences, and PR-AUC. To keep protocol changes accountable, research questions, metrics and gates were written to a file and hashed before each experiment, and every result records that hash, the code commit and the dataset hashes [30]. Seven dated amendments are recorded. Amendments 1 and 5 changed protocols after data-only feasibility counts showed a rule could not be met; Amendments 2–4 each followed a STOP verdict and defined a new pilot and gate; Amendment 6 turned the study into a descriptive benchmark after the fourth STOP and recorded a flaw in the last gate's contrast; and Amendment 7, written after external review and before any run it governs, added the GRU backbone, XGBoost-s, rolling cut-points and operating-point metrics, with every result reported whatever it showed. Every amendment was hashed before the runs it governed, and none changed a recorded verdict.

IV. Evaluation

We first describe the evaluation setup and then answer five questions, one per finding in Sec. I.

A. Evaluation setup

Both backbones are trained with AdamW (learning rate 10^{-3}, weight decay 10^{-4}), cosine annealing, gradient clipping at 1.0, batch size 256 and 15 epochs; the best validation macro-F1 checkpoint is kept (CLAD: best validation AUROC), and every evaluation is repeated with the final-epoch checkpoint. Seeds are 17, 23, 42, 101 and 202, and a seeding regression test runs before every experiment. We report ROC AUC, TPR@1% and PR-AUC per detector and class (mean over seeds, Student-t 95% interval, seed standard deviation); the false-positive threshold is set on each test set's benign windows, so TPR@1% is descriptive and per family (mean over detectors and classes, with a paired bootstrap over classes and seeds, 2,000 resamples). Intervals are descriptive. All 210 final runs (Amendment 7) used one code commit and include the 160 runs of Amendment 6, whose results they reproduce exactly; reports are regenerated from stored result files only. Compute: a CPU-only cloud container (4 vCPUs, Intel Xeon at 2.10 GHz, no GPU); the 210 runs took 10.8 h (185 s per run, both backbones and all detectors). Artifact: https://anonymous.4open.science/r/XXXXXXXX.

TABLE III
Feasible zero-day evaluations per dataset (LOACO folds; natural-novelty classes; rolling class–cut units with feasible cuts out of 4).

Dataset	Grouped	Temporal	Natural	Rolling
D1 CIC-IDS2018	6	0	2	15 (4)
D2 UNSW-NB15	8	8	0	0 (0)
D3 ToN-IoT	6	0	5	15 (3)
D5 Lycos2017	0[a]	0	2	15 (3)

[a] One fold under the purged-block split.

B. Is LOACO feasible on realistic captures?

The first question is whether the standard protocol can even be run once leakage is controlled. Temporal LOACO yields no feasible folds on any of the three day-scheduled captures (D1, D3, D5); only D2, which interleaves attacks in time, supports it (Table III). In D1 and D3 each attack class occupies its own days, so a time-ordered split places a class wholly in training or wholly in test. On D5 the purged split kept 125 of 219 blocks, and only one class (dos_hulk) met both thresholds; web brute force had 199 training windows. Natural novelty, by contrast, offers two to five evaluable classes on every day-scheduled capture, and moving the cut-point multiplies this evidence to 45 units over 19 unique classes, 15 per dataset (D2 has no test-only class at any cut, and D3 no benign validation windows at a 30% cut). This makes the comparison in the next question possible.

C. Does the protocol change the winner?

Given that both protocols can be run on some datasets, the next question is whether they agree. They do not (Table IV). Representation-distance scores lead in five evaluations and benign-only detectors in two. The sharpest reversal is on D3: under LOACO, benign-only and representation scores are close ($+0.022$ [$+0.004$, $+0.047$]), whereas under natural novelty benign-only detectors lead by $+0.376$ [$+0.247$, $+0.500$] and representation scores fall to chance. CLAD never leads; it trails benign-only detectors on both D3 evaluations (-0.123 and -0.075) and is below 0.5 on every seed for D3 LOACO ddos. XGBoost exceeds the leading novelty family on LOACO D2 and natural novelty D3 and D5, and ties it on purged D5, while natural novelty D1 is hard for every family. The GRU

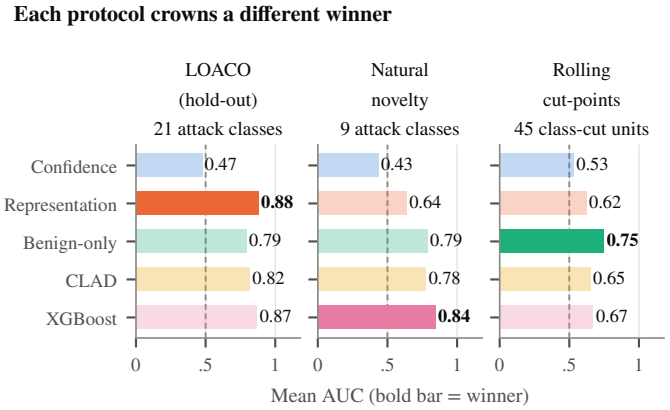


Fig. 2. Mean AUC per detector family pooled over all classes of each protocol (rolling: all 45 class–cut units). The winner (solid bar) differs in each protocol.

TABLE IV
FAMILY MEAN AUC (FIVE SEEDS). BOLD: LEADING NOVELTY FAMILY. CONF. = CONFIDENCE, REPR. = REPRESENTATION, BENIGN = BENIGN-ONLY. CLASSES PER EVALUATION IN PARENTHESES.

Evaluation	Conf.	Repr.	Benign	CLAD	XGB
LOACO D1 (6)	.553	**.729**	.660	.653	.682
LOACO D2 (8)	.439	**.977**	.807	.949	.998
LOACO D3 (6)	.426	.881	**.903**	.780	.862
Purged D5 (1)	.580	**.998**	.833	.972	1.00
Natural D1 (2)	.529	**.623**	.505	.574	.523
Natural D3 (5)	.467	.522	**.898**	.823	.923
Natural D5 (2)	.256	**.940**	.794	.864	.968

backbone selects the same leading family as AHSD in all seven evaluations, so the reversal is not an artifact of one backbone; its absolute values can differ, however (natural novelty D3: GRU representation 0.869 against AHSD's 0.522, still below benign-only detectors at 0.898).

Pooling every class of each protocol shows the effect at a glance (Fig. 2): representation scores win under LOACO, XGBoost under natural novelty, and benign-only detectors over the rolling cut-points. Over the 45 rolling units, benign-only detectors lead (0.748 [0.699, 0.796]) ahead of representation scores (0.623 [0.566, 0.683]; difference $+0.125$ [$+0.048$, $+0.204$]), and XGBoost trails the leading family by -0.076 [-0.154, -0.009]; at a 45% cut on D3 it scores 0.263, below chance. The leader also moves with the dataset (representation on D1 and D5, benign-only on D3) and with the cut fraction (representation, benign-only, CLAD, benign-only from 30% to 75%). A detector chosen on one protocol, dataset or cut-point can therefore be the wrong choice on another.

D. Is one training run enough evidence?

A change of winner could still be a matter of noise, so the third question is how much results vary across seeds. On the first backbone, every stochastic supervised score (the six backbone scores and CLAD) is less stable under natural novelty than under LOACO (Table V). The effect is smaller on the GRU: its representation scores become 2.6× less stable,

TABLE V
MEAN SEED STANDARD DEVIATION OF AUC. LOACO POOLS D1–D3; NATURAL NOVELTY POOLS D1, D3, D5. GRU ROWS: SECOND BACKBONE; XGBOOST-S: SEEDED XGBOOST.

Detector	Family	LOACO	Natural	Ratio
MSP	Conf.	0.061	0.103	1.7×
Energy	Conf.	0.055	0.100	1.8×
$P(\text{attack})^{\dagger}$	Conf.	0.042	0.082	2.0×
Stress	Repr.	0.019	0.110	5.8×
Mahalanobis	Repr.	0.009	0.074	7.9×
kNN	Repr.	0.013	0.215	17.0×
GRU Mahal.	Repr.	0.008	0.022	2.6×
GRU kNN	Repr.	0.019	0.051	2.6×
GRU MSP	Conf.	0.049	0.062	1.3×
GRU $P(\text{attack})$	Conf.	0.027	0.027	1.0×
CLAD	Contr.	0.020	0.058	2.9×
XGBoost-s	Ref.	0.006	0.014	2.4×
Isolation Forest	Benign	0.007	0.008	1.1×
Autoencoder	Benign	0.002	0.017	9.2×

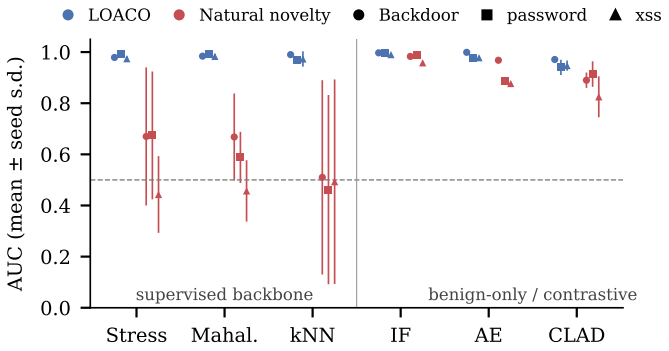


Fig. 3. D3 (ToN-IoT) classes evaluated under both protocols: mean AUC ± seed standard deviation over five seeds. Supervised-backbone scores are tight under LOACO and spread widely under natural novelty; benign-only detectors stay tight. Dashed line: chance.

its confidence scores 1.0–1.3×. Seeding XGBoost (XGBoost-s) raises its seed standard deviation only to 0.014, so the instability belongs to learned representations under shift rather than to seeding as such, and its size depends on the backbone. The seeded benign-only detectors (Isolation Forest, autoencoder) stay below 0.018 seed standard deviation under natural novelty, below every supervised score's minimum of 0.074; OCSVM, PCA and XGBoost are deterministic and carry no evidence about stability. On D3 natural novelty, kNN has a seed standard deviation of 0.33–0.40 AUC on four of five classes, so the same detector can rank first or last depending on the seed. Fig. 3 shows this on the three D3 classes evaluated under both protocols.

E. Which novelty scores are valid?

The confidence family scored worst in Table IV, and Sec. III predicted why. MSP and energy fall below 0.5 wherever the classifier detects unseen attacks confidently (Table VI). On LOACO D2, MSP is below 0.5 on every seed of all eight classes while $P(\text{attack})$ reaches 0.997–1.000 on the same models: the classifier detects the held-out attacks, and MSP reads that confidence as normality. The GRU repeats the

TABLE VI
MEAN AUC OF CONFIDENCE SCORES AND THE REPRESENTATION FAMILY.

Evaluation	MSP	Energy	$P(\text{attack})^{\dagger}$	Repr.
LOACO D2	0.163	0.155	0.999	0.977
LOACO D3	0.269	0.258	0.752	0.881
Natural D3	0.262	0.287	0.853	0.522
Natural D5	0.103	0.090	0.575	0.940

TABLE VII
IN-DISTRIBUTION AUC, GROUPED VS. TEMPORAL SPLIT.

	$P(\text{attack})^{\dagger}$		XGBoost	
Dataset	Grouped	Temporal	Grouped	Temporal
D1	0.812	0.521	0.815	0.547
D2	0.999	0.997	0.998	0.984
D3	0.975	0.664	0.977	0.978
D5	0.831	0.948	0.705	0.990

pattern (LOACO D2: MSP 0.162 vs $P(\text{attack})$ 0.999; natural novelty D3: 0.245 vs 0.918), and in-distribution MSP is 0.065–0.340 on grouped D1–D3 splits. Across all 34 evaluation–backbone pairs, MSP exceeds $P(\text{attack})$ only in five rolling cuts where $P(\text{attack})$ is itself at or near chance (0.345–0.548), i.e. where the classifier misses the novel class. $P(\text{attack})$ falls below 0.5 only on individual classes (D3 LOACO ddos 0.35 and scanning 0.46; D5 portscan 0.16). Representation distance is valid on average, but on natural novelty D3 its interval [0.402, 0.647] includes 0.5. As noted in Sec. III, the $P(\text{attack})$ comparisons are diagnostic; the inversion itself follows from the binary formulation.

F. How do temporal and cross-network shifts behave?

Natural novelty mixes a new attack class with temporal shift, so the last question isolates the shift itself. Its effects are dataset-specific (Table VII). On D1 both models lose 0.27–0.29 AUC under the temporal split; on D2 neither changes by more than 0.014; and on D5 both improve. Only on D3 does the AHSD backbone degrade (0.664, seed s.d. 0.202) while XGBoost holds 0.978; the GRU degrades less (0.981→0.857), so the size of this drop depends on the backbone. Across 108 zero-shot transfer cells (12 detectors × 9 source–target pairs among D1–D4), the median AUC is 0.500 and 67% fall in [0.3, 0.6] (Table XVII). Checkpoint choice moves representation and CLAD family means by at most 0.073 and confidence scores by up to 0.205 (D5, 26 validation attack windows).

G. Does AUC reflect detection at an operating point?

An operator does not experience AUC but the share of attacks caught at a false-alarm rate the security team can absorb. The two disagree sharply (Fig. 4). Under LOACO D2 the best families catch 87–99% of held-out windows at 1% false-positive rate. Under natural novelty D3, benign-only detectors reach an AUC of 0.90 but catch 12%, and XGBoost 38%. Over the 45 rolling units no family exceeds 18% (XGBoost 0.18; benign-only 0.09 at AUC 0.75), and of the 333 detector–class points with AUC ≥ 0.75 under natural

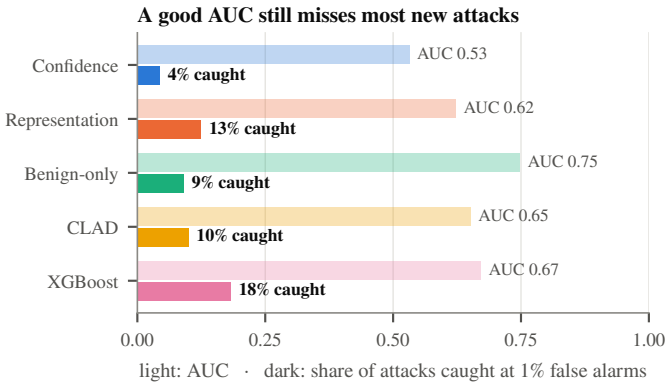


Fig. 4. Rolling cut-points (45 units): each family's AUC (light bar) against the share of novel attack windows it catches at 1% false-positive rate (dark bar).

novelty, 184 catch fewer than 20% of attack windows (median 0.15): novel attacks overlap the benign score tail.

H. Pre-registered negative results

Before the benchmark became descriptive, four confirmatory hypotheses were tested against frozen gates, and all four failed (Appendix B): a spectral model of per-class detectability (pooled ρ=0.326, below 0.6); attack absorption by an adaptive equilibrium under supervised training (not reproduced; detectability peaks at a small adaptation rate); benign-drift re-anchoring (0 of 9 detectors met the improvement criterion); and benign-only superiority confined to natural novelty. The last gate's supervised family included MSP and energy, which Sec. IV-E shows are invalid; the verdict stands as recorded.

V. DISCUSSION

Main findings. The five questions give a consistent answer to the paper's operational question. On realistic captures, the standard zero-day test is often impossible; where it can be run, it can crown a different winner than natural novelty, and the winner moves again with the cut-point; single-run results can understate run-to-run variation by up to 17×; and an AUC that looks adequate can mean that most novel attacks go unnoticed at an affordable false-alarm rate. A zero-day claim backed only by single-run LOACO on a random split therefore says little about how a detector will behave once deployed.

Interpretation. The protocols disagree because they change different things. LOACO keeps the benign traffic of the test period in training, so supervised representations place a held-out class far from benign traffic and distance scores excel (0.977 on D2). Under natural novelty the benign traffic itself shifts, and representations shaped by known attacks lose their reference point, which explains both the drop to chance and the seed dependence on D3; how far they fall depends on the backbone (GRU: 0.869). Benign-only detectors never depend on which attacks were trained on, which explains their stability, but not their accuracy: they lead on D3 natural novelty yet trail by 0.17 AUC on D2 LOACO. The strength of XGBoost on fixed evaluations is consistent with evidence that

tree ensembles remain hard to beat on tabular data [31], and flattened flow windows are tabular data; its collapse at some cut-points shows that a supervised boundary can also place a new class on the wrong side.

What works: the HoNE acceptance protocol. These mechanisms suggest what evidence does predict deployment behaviour, and the HoNE acceptance protocol asks an operator to require it before trusting a learned NIDS to detect zero-day attacks:

1) **Leakage controls stated:** time-block grouping before windowing, window deduplication, and purging for short captures.
2) **Natural novelty reported** wherever timestamps allow, next to LOACO, with an explicit statement when either is infeasible.
3) **Five or more seeds** with per-class seed standard deviation, not only means.
4) **No MSP or energy** as novelty scores for a binary detector; use $P(\text{attack})$ or a representation distance validated under natural novelty.
5) **Two baselines**: a benign-only detector and a gradient-boosted tree. In our benchmark XGBoost alone matched or exceeded the best novelty family in four of seven fixed evaluations, XGBoost or the benign-only family matched or exceeded every other family in five of seven, and benign-only detectors led over the rolling cut-points.
6) **Operating-point metrics**: TPR at the false-positive rate the security team can absorb, not only AUC.
7) **Checkpoint sensitivity** when validation sets are small.
8) **Pre-registration or a full amendment log**, including failed hypotheses.

These checks cost little: each of our 210 final runs, covering two backbones and all detectors, took about three minutes on four CPU cores.

Remaining limitations. The protocol's evidence has limits that bound how far it generalizes. Both backbones are recurrent, so attention-based models may behave differently, and the size of the seed and temporal effects already differs between them. D1 and D5 contribute two natural-novelty classes each in the primary split; the rolling cuts add 45 units, but units of one class at several cuts share test windows and are not independent. D5's validation set holds only 26 attack windows. The 1-hour grouping, any-attack labelling, 200-window threshold and 5:1 test ratio were fixed in advance but are not the only reasonable choices; $P(\text{attack})$ was added post hoc; and CLAD was reimplemented from its paper, although it reaches 0.972 on purged D5, close to the 0.997 its authors report on the same dataset [1]. Finally, all datasets come from testbeds, and natural novelty follows the capture schedule, which is closer to deployment than LOACO but is not live traffic.

Broader implications. Within these limits, the findings matter beyond this benchmark. For network management, the acceptance protocol can be written into procurement: a vendor can be asked to report results on HoNE's released splits, under both protocols and over several seeds. For research, the results argue that zero-day claims should be accompanied by protocol-paired, multi-seed evidence, and that negative pre-registered results deserve publication, because they show where intuitive hypotheses about detector behaviour fail.

VI. Conclusion

Operators must decide whether to trust learned NIDS on attacks that did not exist at training time, yet the evidence offered for such claims is usually a single LOACO run on a randomly split benchmark. We introduced HoNE, a pre-registered, leakage-controlled benchmark that scores every detector under both LOACO and natural temporal novelty, and applied it to five datasets, 12 detectors on two backbones and five seeds. Temporal LOACO proved infeasible on every day-scheduled capture; protocols, datasets and cut-points crowned different winners, on both backbones; natural novelty raised seed variability by up to $17\times$; max-softmax and energy proved invalid for binary detectors; and AUCs up to 0.75 meant at most 18% of novel attacks caught at 1% false-positive rate. How zero-day detection is evaluated decides which detector looks best, so operators should ask for natural-novelty results at several cut-points, multiple seeds, valid scores, operating-point metrics and simple baselines before trusting a zero-day claim. We release HoNE's splits, amendments and code so that any detector can be evaluated under both protocols.

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TABLE VIII
POSITIONING. – = NOT ASSESSED IN THIS COMPARISON.

Work	Zero-day protocol	Cross-dataset	Seed var.	Pre-reg.
[15]	Zero-shot hold-out	No	–	No
[14]	Simulated unknowns	No	–	No
[16]	No	Yes	–	No
[1]	LOACO	No	–	No
[2]	Leave-one-type-out	1 pair	5 seeds	No
HoNE	LOACO, natural, rolling, purged	9 pairs	5 seeds × 2 backb.	Yes

TABLE IX
PRE-REGISTERED GATES. ALL FOUR RETURNED STOP.

Gate	Hypothesis	Frozen criterion	Observed
P1	Spectral filter model predicts per-class detectability	Pooled $\rho \geq 0.6$ and better than Wasserstein	ρ=0.326; difference CI [−0.578, 1.286]
G1	Adaptive equilibrium absorbs attacks under supervised training	$\rho(\gamma, d') \leq -0.8$ on ≥2 of 3 datasets	−0.143, −0.143, 0.000
G2	Benign drift causes inversion; re-anchoring fixes it	≥5 of 9 detectors pass each criterion	4/9, 5/9, 0/9
G3	Benign-only beats supervised only under natural novelty	$B-S \geq 0.15$ natural; ≤ 0.05 LOACO	+0.191 (D5); +0.269 (D3 LOACO)

APPENDIX A
EXTENDED RELATED WORK

Evaluation pitfalls. Arp et al. identified ten pitfalls in security ML, including data snooping and spurious correlations [3]. Apruzzese et al. argued that ML-NIDS research rarely supports operational assessment [8]. Sommer and Paxson explained why anomaly detection is harder outside a closed world [7].

Zero-day and open-set detection. Kitsune detects anomalies with an ensemble of autoencoders trained on benign traffic [32]. Wilkie et al. (2026) learn benign structure from benign and known-malicious samples with a contrastive loss [1]. Jodayree et al. (2026) combine a protocol-aware transformer with energy scoring and extreme-value calibration [2]. Both use class hold-out.

Cross-domain generalization. Beyond [16]–[18], a 2026 systematic review surveys deep IoT intrusion detection and its evaluation practice [33].

Sequence and neuromorphic NIDS. FlowTransformer provides a framework for transformer-based flow NIDS [34]. Wang et al. built a convolutional spiking detector [35], and Knapp et al. evaluated bare spiking networks against conventional and recurrent networks [36]; both report in-distribution accuracy.

Drift and adversaries. At NDSS 2026, a reinforcement-learning attacker evaded six ML traffic detectors without model access [37], and Entente addressed cross-silo detection on log graphs with federated learning [38]. HOIDS combines online learning with drift awareness [39], and CADE detects and explains drifting samples [40].

APPENDIX B
PRE-REGISTERED GATES AND AMENDMENTS

Amendment log. A1: LOACO moved to grouped_random; D1 classes defined as families; D4 made a transfer target. A2: spectral predictor demoted to secondary; D1 natural novelty dropped (validation empty); new gate G1. A3: G1 negative; train-tail validation introduced, which restored D1 natural novelty; new gate G2. A4: G2 negative; Lycos2017 and CLAD added; new gate G3. A5: purged-block protocol for D5 with a pre-declared fallback; checkpoint sensitivity. A6: descriptive benchmark; gate flaw recorded; $P(\text{attack})$ added. A7 (after external review, before any run it governs): GRU backbone, seeded XGBoost-s, rolling cut-points and operating-point metrics; full rerun of 210 runs at one code commit.

TABLE X
PER-CLASS AUC, LOACO D1 (CIC-IDS2018 FAMILIES). MEAN OF FIVE SEEDS; S.D. = MEAN SEED STANDARD DEVIATION OVER CLASSES.

Detector	Bot	Brute F.	DDoS	DoS	Infil.	Web	s.d.
MSP	0.67	0.50	0.51	0.28	0.49	0.66	0.093
Energy	0.58	0.48	0.50	0.19	0.49	0.64	0.072
P(att.)†	0.73	0.71	0.51	0.84	0.52	0.66	0.103
Stress	0.82	0.81	0.56	0.94	0.60	0.57	0.032
Mahal.	0.83	0.92	0.61	0.96	0.53	0.57	0.023
kNN	0.82	0.87	0.57	0.92	0.56	0.67	0.028
IF	0.54	0.73	0.55	0.87	0.52	0.57	0.013
OCSVM	0.30	0.39	0.88	0.80	0.47	0.62	0.000
PCA	0.60	0.72	0.98	0.90	0.52	0.76	0.000
AE	0.53	0.62	0.92	0.86	0.48	0.71	0.002
CLAD	0.66	0.58	0.67	0.94	0.50	0.57	0.041
XGBoost	0.73	0.81	0.58	0.83	0.54	0.59	0.000

TABLE XI
PER-CLASS AUC, LOACO D2 (UNSW-NB15).

Detector	Anal.	Backd.	DoS	Expl.	Fuzz.	Gen.	Recon.	Shellc.	s.d.
MSP	0.06	0.06	0.10	0.15	0.08	0.34	0.27	0.24	0.021
Energy	0.05	0.06	0.09	0.13	0.07	0.33	0.27	0.23	0.025
P(att.)†	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.000
Stress	0.98	0.94	0.95	0.92	0.96	0.92	0.90	0.91	0.016
Mahal.	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.001
kNN	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.000
IF	0.66	0.64	0.66	0.69	0.72	0.74	0.66	0.73	0.007
OCSVM	0.81	0.71	0.74	0.76	0.82	0.86	0.74	0.81	0.000
PCA	0.93	0.84	0.86	0.86	0.92	0.94	0.87	0.90	0.000
AE	0.91	0.83	0.84	0.85	0.90	0.92	0.84	0.89	0.002
CLAD	0.96	0.94	0.94	0.94	0.96	0.95	0.95	0.94	0.004
XGBoost	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.000

APPENDIX C
PER-CLASS AND TRANSFER RESULTS

TABLE XII
PER-CLASS AUC, LOACO D3 (TON-IOT).

Detector	Backd.	ddos	inject.	passw.	scan.	xss	s.d.
MSP	0.24	0.30	0.20	0.18	0.44	0.26	0.083
Energy	0.23	0.30	0.16	0.13	0.43	0.30	0.078
P(att.)†	0.94	0.35	0.91	0.86	0.46	0.98	0.036
Stress	0.98	0.76	0.99	0.99	0.59	0.97	0.010
Mahal.	0.98	0.91	0.99	0.99	0.57	0.98	0.007
kNN	0.99	0.70	0.98	0.97	0.52	0.97	0.014
IF	1.00	0.89	1.00	1.00	0.54	0.99	0.002
OCSVM	1.00	0.86	0.99	0.98	0.58	0.98	0.000
PCA	1.00	0.88	0.98	0.98	0.62	0.98	0.000
AE	1.00	0.87	0.98	0.98	0.63	0.98	0.001
CLAD	0.97	0.37	0.95	0.94	0.50	0.95	0.019
XGBoost	0.97	0.69	1.00	1.00	0.53	1.00	0.000

TABLE XIII
PER-CLASS AUC, NATURAL NOVELTY D1.

Detector	Bot	Infil.	s.d.
MSP	0.59	0.45	0.054
Energy	0.59	0.45	0.054
P(att.)†	0.59	0.50	0.055
Stress	0.67	0.57	0.015
Mahal.	0.69	0.57	0.015
kNN	0.69	0.55	0.020
IF	0.51	0.48	0.010
OCSVM	0.38	0.50	0.000
PCA	0.59	0.53	0.000
AE	0.55	0.50	0.001
CLAD	0.63	0.52	0.029
XGBoost	0.51	0.54	0.000

TABLE XIV
PER-CLASS AUC, NATURAL NOVELTY D3.

Detector	Backd.	mitm	passw.	ransom.	xss	s.d.
MSP	0.13	0.52	0.19	0.34	0.14	0.116
Energy	0.16	0.53	0.21	0.38	0.16	0.117
P(att.)†	0.93	0.67	0.91	0.85	0.90	0.080
Stress	0.67	0.49	0.67	0.42	0.44	0.191
Mahal.	0.67	0.54	0.59	0.38	0.46	0.126
kNN	0.51	0.57	0.46	0.46	0.49	0.323
IF	0.98	0.70	0.99	0.95	0.96	0.003
OCSVM	0.98	0.80	0.89	0.92	0.89	0.000
PCA	0.93	0.84	0.88	0.91	0.86	0.000
AE	0.97	0.83	0.89	0.91	0.88	0.002
CLAD	0.89	0.64	0.91	0.84	0.83	0.055
XGBoost	0.96	0.77	0.99	0.92	0.99	0.000

TABLE XV
PER-CLASS AUC, NATURAL NOVELTY D5 (LYCOS2017).

Detector	ddos	portscan	s.d.
MSP	0.05	0.16	0.118
Energy	0.05	0.13	0.105
P(att.)†	0.99	0.16	0.115
Stress	0.99	0.98	0.003
Mahal.	0.99	0.98	0.003
kNN	0.95	0.75	0.139
IF	0.78	0.99	0.018
OCSVM	0.49	0.98	0.000
PCA	0.86	0.81	0.000
AE	0.86	0.58	0.071
CLAD	0.99	0.74	0.094
XGBoost	1.00	0.94	0.000

TABLE XVI
AUC, PURGED-BLOCK LOACO D5 (SINGLE FOLD).

Detector	DoS Hulk	s.d.
MSP	0.40	0.338
Energy	0.34	0.308
P(att.)†	1.00	0.001
Stress	1.00	0.002
Mahal.	1.00	0.002
kNN	1.00	0.003
IF	0.96	0.011
OCSVM	0.51	0.000
PCA	0.89	0.000
AE	0.97	0.004
CLAD	0.97	0.014
XGBoost	1.00	0.000

TABLE XVII

ZERO-SHOT TRANSFER AUC (MEAN OF FIVE SEEDS). ROWS: DETECTOR; COLUMN GROUPS: SOURCE DATASET; COLUMNS: TARGET. DIAGONAL CELLS (IN-DOMAIN) IN GREY.

	Source D1				Source D2				Source D3			
Detector	D1	D2	D3	D4	D1	D2	D3	D4	D1	D2	D3	D4
MSP	0.51	0.48	0.34	0.58	0.50	0.08	0.33	0.69	0.44	0.50	0.36	0.23
Energy	0.51	0.51	0.34	0.46	0.50	0.08	0.33	0.68	0.45	0.71	0.32	0.31
P(att.)[†]	0.52	0.55	0.57	0.36	0.41	1.00	0.28	0.29	0.56	0.50	0.66	0.48
Stress	0.64	0.36	0.53	0.39	0.36	0.95	0.18	0.16	0.54	0.31	0.36	0.71
Mahal.	0.66	0.44	0.43	0.54	0.38	1.00	0.16	0.20	0.55	0.28	0.40	0.61
kNN	0.63	0.48	0.44	0.44	0.41	0.99	0.11	0.57	0.59	0.38	0.62	0.67
IF	0.49	0.54	0.44	0.70	0.28	0.70	0.04	0.73	0.55	0.51	0.95	0.80
OCSVM	0.39	0.60	0.03	0.54	0.50	0.74	0.57	0.70	0.59	0.50	0.89	0.64
PCA	0.56	0.56	0.67	0.60	0.50	0.80	0.48	0.68	0.55	0.61	0.87	0.57
AE	0.52	0.62	0.47	0.59	0.50	0.80	0.58	0.71	0.58	0.60	0.88	0.58
CLAD	0.57	0.42	0.88	0.35	0.47	0.84	0.29	0.34	0.56	0.45	0.81	0.64
XGBoost	0.55	0.59	0.83	0.64	0.52	0.98	0.46	0.43	0.50	0.44	0.98	0.71